

CO₂ capture with polyamine-based protic ionic liquid functionalized mesoporous silica

Wei Zhang^{a,b}, Erhao Gao^{a,b}, Yu Li^{a,b}, Matthew T. Bernards^c, Yi He^{a,d,*}, Yao Shi^{a,b,*}

^a College of Chemical and Biological Engineering, Zhejiang University, Hangzhou, 310027, PR China

^b Key Laboratory of Biomass Chemical Engineering of Ministry of Education, Zhejiang University, Hangzhou, PR China

^c Department of Chemical and Materials Engineering, University of Idaho, Moscow, 83844, USA

^d Department of Chemical Engineering, University of Washington, Seattle, Washington 98195, USA

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ABSTRACT

A mesoporous silica supported polyamine-based protic ionic liquid (PIL) sorbent has been prepared by impregnating tetraethylenepentammonium nitrate [TEPA][NO₃] onto the surface of SBA-15. This innovative hybrid adsorbent exhibits an excellent CO₂ adsorption rate in the rate-controlling step of $147 \times 10^{-3} \text{ mmol g}^{-1} \text{ s}^{-0.5}$, at least threefold better than other IL-functionalized and amine-modified support systems. Moreover, the adsorption capacity is enhanced by up to 2.15 mmol/g for the adsorbent at 333 K and 0.15 bar, which represents excellent performance under low CO₂ partial pressures. This adsorption performance can be ascribed to the synergistic effect of strong affinity between the PILs and CO₂ and large surface area of the support which favors the diffusion of the PILs and CO₂ molecules in its pores. In addition, density functional theory (DFT) calculations further reveal that CO₂ preferentially reacts with the secondary amine –N(2)H in [TEPA][NO₃] forming carbamate due to the high binding energy. The cyclic adsorption/regeneration experiment shows that nearly 90% of original CO₂ adsorption capacity of the adsorbent is restored by N₂ steam desorption at 373 K. Therefore this novel silica supported PIL system shows promise for CO₂ removal and recovery applications.

1. Introduction

It is incontestable that global warming and anthropogenic climate change are associated with increasing levels of atmospheric CO₂ due to the large-scale burning of fossil fuels which started during the industrial revolution [1]. Scripps Institute of Oceanography measured atmospheric CO₂ concentrations increasing from 315 ppm in 1958 to 391 ppm in 2011, and close to 398 ppm in 2014 [2]. While the full use of clean energy seems to be the ultimate solution, carbon capture and sequestration (CCS) technologies represent a viable short-term strategy for emissions mitigation [3]. Currently, there are three types of technologies to capture CO₂ from point sources, pre-combustion capture, post-combustion capture, and oxyfuel combustion [4]. Compared with the other two technologies, the post-combustion capture method is highly competitive based on cost and ease of retrofitting existing facilities [5].

Numerous materials are being developed for CO₂ capture both in the laboratory and on an industrial scale. There are three main materials including solvents, solid sorbents, and membranes [6]. Among them, aqueous amine solvents are the benchmark technology for CO₂

absorption [7,8]. However, they suffer from serious defects including the high energy penalty associated with the high specific heat capacity of water and the significant operational costs that result from corrosion, thermo-oxidative degradation, and volatility of the amine solvents [9]. In order to circumvent these challenges, ionic liquids (ILs) have been proposed as promising alternatives due to their unique physicochemical properties such as low corrosiveness, high thermal stability, negligible vapor pressure, and low melting point [10,11]. Blanchard et al. reported that a large quantity of CO₂ could be absorbed in imidazolium-based ILs [12]. Their study stimulated a considerable explosion of interest and research activities in this field [13]. To improve the CO₂ absorption capacity, Bates et al. synthesized a task-specific IL by introducing a primary amine moiety into alkylimidazolium cations, and the molar uptake of CO₂ per mole for these ILs approached the theoretical maximum of 0.5 [14]. To go beyond this limitation, Gurkan et al. developed several amino acid-based ILs using proline and methionine as anions, and the CO₂ molar absorption capacity achieved one CO₂ per amine [15]. However, the high cost and tedious synthetic procedures of the above-mentioned ILs still limit their practical applications on an industrial scale.

* Corresponding authors at: College of Chemical and Biological Engineering, Zhejiang University, Hangzhou, 310027, PR China.

E-mail addresses: yihezj@zju.edu.cn (Y. He), shiyao@zju.edu.cn (Y. Shi).

Compared to conventional ILs, protic ionic liquids (PILs) not only have the advantages of low corrosiveness, high thermal stability, negligible vapor pressure, and low melting point, but also have the merits of low-cost, easy synthesis, high CO₂ absorption capacity and selectivity. Paul Walden first reported a room-temperature protic ionic liquid, namely ethylammonium nitrate (EAN) with a low melting point of 285 K, negligible vapor pressure, and high thermal stability [16]. EAN was synthesized by the neutralization of low-priced ethylamine with concentrated nitric acid. After that, Ilioudis et al. synthesized a polyamine-based PIL using diethylenetriamine and nitric acid by a one-step method [17]. Doyle and coworkers evaluated its CO₂ absorption capability and it showed a high affinity for CO₂ [18]. Hu et al. developed a series of polyamine-based PILs with the same triethylenetetramine (TETA) cation and various inorganic acid anions, and the CO₂ absorption capacity of these ILs followed the order of [NO₃] > [BF₄] > [SO₄] or [HSO₄]. In addition, the 40 wt% [TETA][NO₃] showed the best capacity with 1.49 mol CO₂ per mol PIL at 288 K and 1 bar [19].

Although the polyamine-based PILs have many merits, their CO₂ absorption rates are quite low due to their high viscosities. Loading PILs onto a porous material can be an effective approach to tackle this challenge. This approach has been applied to address similar issues with ILs when they were used for CO₂ capture [20]. There are several widely studied supporting materials such as mesoporous silicon-based materials [21,22], metal-organic frameworks (MOFs) [23], nanoporous polymers, [24] and membranes [25]. Among these, the mesoporous molecular sieve SBA-15 attracts lots of attention due to its highly ordered 2D hexagonal (p6mm) pore path, uniform pore size, high hydrothermal stability, large specific surface area, and good compatibility with various ILs. Ruckart et al. [26] synthesized six task-specific ionic liquids functionalized ordered mesoporous silica SBA-15 and amorphous porous support BPL activated carbon, and investigated the effect of the supports on CO₂ adsorption capacity of the hybrid sorbents. The results demonstrated that the SBA-15 hybrid sorbents exhibited better CO₂ adsorption capacity and stability. Arellano et al. [27] developed a number of SBA-15 supported zinc-functionalized IL sorbents, and the S15-50 which contained 50 wt% IL exhibited a high CO₂ uptake capacity of 4.7 mmol/g and a rapid adsorption rate. Cheng et al. [28] synthesized a series of SBA-15 hybrid sorbents by introducing trihexyl (tetradecyl)phosphonium 1,2,4-triazole into four kinds of SBA-15 which had different pore structures. The CO₂ adsorption rate of SBA-15(4.3)-50%IL was faster than the other three kinds of hybrid sorbents, because it had the smallest pore size and highest total pore cross-sectional area.

In order to obtain an adsorbent with both high CO₂ adsorption rate and large adsorption capacity, while possessing desirable features such as low cost, we developed a series of mesoporous silica SBA-15 supported polyamine-based PIL sorbents. The polyamine-based PILs were synthesized by one-step acid-base neutralization of low-priced tetraethylenepentamine (TEPA) with nitric acid. The hybrids between the PIL and SBA-15 were synthesized in different proportions through an impregnation approach. In addition, the structural properties of the hybrid sorbents were investigated by various physical characterization methods. Breakthrough experiments were used to analyze their dynamic CO₂ adsorption capacities and the Weber-Morris intra-particle diffusion model was carried out to discuss CO₂ adsorption kinetics. Moreover, the adsorption mechanism of CO₂ on the sorbent was examined by density functional theory (DFT) calculations.

2. Experimental and simulation details

2.1. Materials

Amphiphilic triblock copolymer Pluronic P123 (Mw 5800) was purchased from Sigma Aldrich. Tetraethyl orthosilicate (TEOS 99.7%), hydrochloric acid (HCl, 36–38%), nitric acid (HNO₃, 65.0–68.0%), and

absolute ethyl alcohol (EtOH, 99.7%) were purchased from Sinopharm Chemical Reagent Co., Ltd., Shanghai, China. Tetraethylenepentamine (TEPA, 99.5%) was obtained from Shanghai Aladdin biochemical technology co., LTD. Carbon dioxide (99.999%), nitrogen (99.999%), hydrogen (99.999%), and mixed gases of CO₂/N₂ = 15/75 were purchased from Jingong (China). All chemicals were commercially available and used directly without further purification.

2.2. Synthesis of SBA-15 and polyamine-based PILs

SBA-15 was synthesized by a previous method [29] with minor modifications. The Pluronic P123 solution was obtained by mixing Pluronic P123 (20.0 g), HCl (125 mL), and deionized water (625.0 mL) with a magnetic stirrer at 313 K. When P123 is completely dissolved, TEOS (42.5 mL) was added in the solution and stirred for 20 h, then the mixed solution was transferred to two 500 ml Teflon-lined autoclaves. The autoclaves were heated to 393 K within 2 h and kept at 393 K for 24 h. The SBA-15 precursors were obtained by filtration, washed thoroughly with water, and dried at 373 K for 12 h. Finally, the SBA-15 precursors were calcined at 773 K for 8 h in a tube furnace in air, yielding surfactant-free SBA-15.

The tetraethylenepentammonium nitrate ([TEPA][NO₃]) was synthesized by one-step acid-base neutralization of TEPA with HNO₃. Aqueous nitrate solution (0.5 mol, 40 wt%) was added dropwise into an aqueous TEPA solution (0.5 mol, 60 wt%) in an ice-water bath with stirring and N₂ protection. To ensure complete proton exchange, the mixture was continually stirred at 293 K for an additional 4 h. Then, the remaining water in the mixture was removed by heating at 373 K for 24 h under vacuum conditions. The resulting product was orange [TEPA][NO₃]. The water content in [TEPA][NO₃] was less than 0.5 wt % monitored via Karl Fischer titration.

2.3. Preparation of PIL functionalized SBA-15

The hybrid sorbents were prepared via the wet impregnation method. Specifically, a desired amount of [TEPA][NO₃] was dissolved in 30 g of ethanol under stirring. After complete dissolution, 1 g of SBA-15 was added to the solution and stirred for 6 h at room temperature. Then, the ethanol in the mixture was evaporated at 363 K for 2 h in open air. The residue was dried at 373 K under vacuum for 12 h. The obtained samples were denoted as S15-x[TEPA], where [TEPA] represents [TEPA][NO₃] and x refers to the mass ratio of the PIL to the support.

2.4. Characterization of sorbents

The surface area, pore volume, and pore diameter distribution of samples were characterized with nitrogen physisorption measurements (ASAP 2460, Micromeritics Inc., USA) at 77.4 K. Prior to nitrogen adsorption-desorption measurements, the samples were degassed at 403 K for 12 h. The specific surface area (SSA) of sorbents was calculated with the Brunauer-Emmett-Teller (BET) model over a relative pressure (P/P₀) range of 0.05–0.30. The Barrett-Joyner-Halenda (BJH) method with desorption branch was used to analyze pore size distribution. Thermogravimetric analysis (TGA, SDT Q600, TA Instruments, Inc., New Castle, DE) was carried out to analyze the thermal stabilities of samples over the temperature range of 303–773 K with a 10 K/min heating rate and N₂ atmosphere. The elemental analyzer (Flash EA 1112, ThermoFinnigan, Italy) was used to analyze the weight percentage of elements (C, H and N) in the composites. The crystal phase of samples was characterized by small-angle x-ray scattering (SAXS) (SAXS/WAXS SYSTEM, XENOCs Co., Ltd., France) using Cu K α radiation (λ = 0.15418 nm) at 50 kV and 0.6 mA over the 2θ range of 0.5°–3.5°. FT-IR spectroscopy (NICOLET 6700, thermal scientific, USA) was carried out between 4000 and 400 cm^{−1} to identify the surface functional groups of the sorbents.

2.5. Adsorption experiments

The CO₂ adsorption isotherms of samples at different temperatures were characterized with CO₂ physisorption measurements (ASAP 2460, Micromeritics Inc., USA). The samples were degassed at 373 K for 12 h under vacuum prior to CO₂ adsorption analyses.

The dynamic CO₂ adsorption capacities of sorbents were characterized by breakthrough experiments. The breakthrough curves for CO₂ adsorption to samples at different temperatures were measured with fixed bed equipment under ambient pressure. In a typical experimental procedure, ~500 mg of sample was added into a glass column with a length of 14 cm and an inner diameter of 6 mm, supported by quartz wool, and the packing height of the sample was about 6.5 cm. The temperatures were controlled with a heating jacket. Prior to CO₂ adsorption analyses, the samples were degassed under a nitrogen flow of 50 mL/min at 423 K for 1.5 h and cooled down to the test temperature. Then the flow was switched to a mixture gas of 15 vol % CO₂ and 85 vol % N₂ at a flow rate of 20 mL/min. The CO₂ concentration of the outlet gas was detected by a gas chromatograph with a TCD detector. The CO₂ uptake capacities of samples were described based on changes in CO₂ concentration of the outlet gas over time. The CO₂ adsorption capacities were calculated by Eq. (1) as follows:

$$q_t = \frac{1}{M} \left[\int_0^t Q \frac{C_0 - C_t}{1 - C_t} dt \right] \frac{T_0}{T} \frac{1}{V_m} \quad (1)$$

where q_t is adsorption amount of CO₂ (mmol/g); M refers to the weight of sorbent (g); Q refers to the mixture gas flow rate (mL/min); T_0 is 273 K; T is operating temperature (K); V_m is 22.4 mL/mmol; C_0 and C_t are CO₂ content (vol %) of the influent and effluent, respectively.

2.6. Density functional theory calculations

Density functional theory (DFT) simulations using DMol³ code in Material Studio 2017 R2 were conducted to optimize geometry and calculate the binding energies (BE) between different amine groups of PILs and CO₂. The calculations were carried out with Becke's three-parameter exchange functional and gradient-corrected functional of Lee, Yang, and Parr (B3LYP) [30,31], using double numerical plus polarization (DNP) basis sets. The Grimme method [32] was utilized to describe the long-range dispersion interactions. The BE of PILs–CO₂ complexes were obtained with the following equation:

$$BE = E[PILs - CO_2] - E[PILs] - E[CO_2] \quad (2)$$

Where, $E[PILs - CO_2]$ is the total energy of optimized PILs–CO₂ complexes; $E[PILs]$ and $E[CO_2]$ are the total energies of optimized PILs and CO₂ molecules, respectively.

3. Results and discussion

3.1. Characterization of hybrid sorbents

3.1.1. Specific surface area and pore diameter analysis

The impregnation treatment used for the preparation of sorbents may significantly affect their physical properties, which are vital for CO₂ adsorption [28,33,34]. The addition of the PILs result in an expected reduction in specific surface area, pore volume, and average pore size indicative of pore filling. Fig. 1 shows N₂ adsorption-desorption isotherms and specific surface area results for the sorbents as a function of the PIL loading. These results demonstrate that each sorbent exhibits an IV-type isotherm as well as an H1-shaped hysteresis loop in the relative pressure range of $0.7 \leq P/P_0 \leq 0.9$, indicating a mesoporous structure with negligible micropores. In addition, Fig. 1b and Table 1 exhibit results demonstrating that the SSA, pore volume, and average pore size of the sorbents are decreased as the PIL loading increases and the SSA has good correlation with the PIL loading

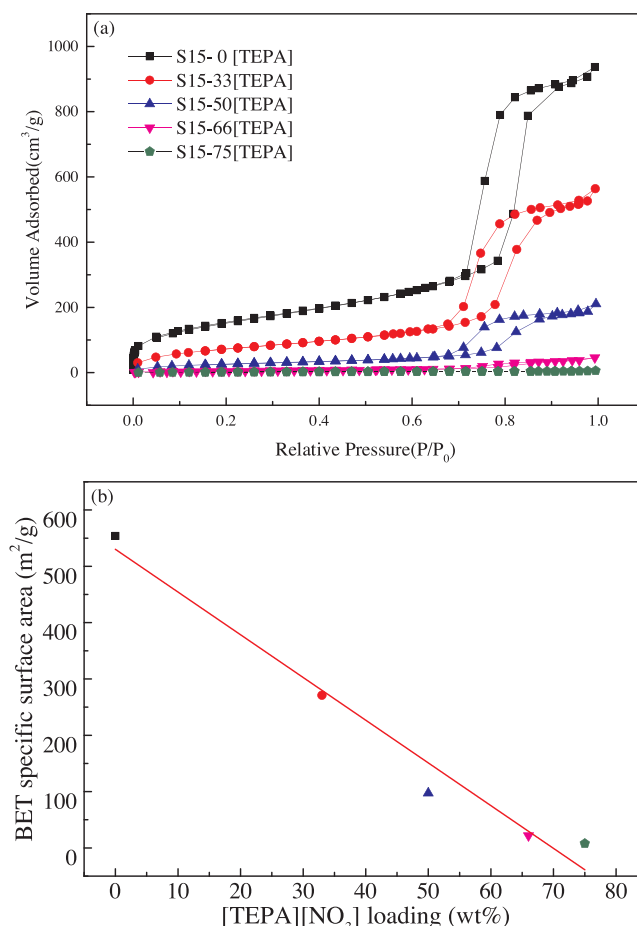


Fig. 1. N₂ adsorption-desorption isotherms at 77 K (a) and specific surface area of S15-x[TEPA] as a function of the PIL loading (b). The SSA and pore volume of S15-75[TEPA] are too small, thus the physical properties of sorbents with more PIL loading were not examined.

Table 1
Physical properties of S15-x[TEPA].

Sample	S _{BET} (m ² /g)	V _t (cm ³ /g)	d _{BH} (nm)
S15- 0 [TEPA]	553.93	1.44	5.26
S15-33[TEPA]	271.02	0.86	5.06
S15-50[TEPA]	97.29	0.32	3.71
S15-66[TEPA]	21.90	0.09	3.22
S15-75[TEPA]	7.82	0.008	3.53

($R^2 > 0.96$). Further, after impregnating the PILs into SBA-15, the porous physicochemical properties of the support are negatively impacted, leading to a decrease in the CO₂ adsorption rate in the initial stage.

3.1.2. Crystal phase analysis

To analyze the effect of PILs on the crystal phase of SBA-15, small angle X-rays diffraction was used to obtain the structural properties of the hybrid sorbents. Fig. 2 shows that almost all samples have 2θ diffraction peaks at 0.83°, 1.44° and 1.67°, which correspond to the (100), (110), and (200) crystallographic planes in 2D hexagonal (p6mm) mesoporous silica [29,35]. It also shows that the intensities of the three diffraction peaks significantly decrease when the PIL content exceeds 50%. This indicates that when excess PILs reside in SBA-15 it results in a decrease in the order of the 2D hexagonal structure. Additionally, the diffraction peak representative of the (100) plane of S15-33[TEPA] and S15-50[TEPA] show a slight deviation toward a larger angle, which can

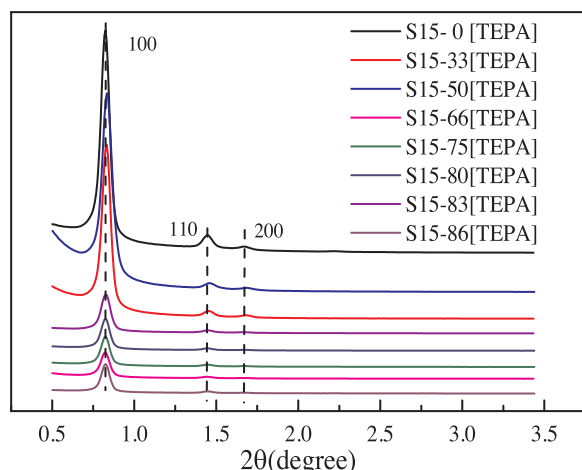


Fig. 2. SAXS spectra and crystallographic planes distribution of S15-x[TEPA].

be explained as pore swelling that results from the PIL treatment.

3.1.3. Thermogravimetric analysis

The thermostability of the sorbents, accessed by thermogravimetric analysis, is another important property for evaluating its potential for CO₂ capture. The results demonstrate that the thermostability of S15-66[TEPA] is comparable to other IL-functional molecular sieves [36] and is better than TEPA-modified porous materials [34,37]. Fig. 3 shows that the weight loss of SBA-15 exhibits two stages: the first weight loss, below 373 K, is 2.53% which can be ascribed to the evaporation of physisorbed moisture; the second region, from 373 K to 773 K, is 4.25% due to the dehydration of hydroxyl groups on the wall surface and the decomposition of residual template P123. In the case of S15-66[TEPA], the weight loss is divided into three stages. In the first region, the weight loss is 5.74% and this is ascribed to moisture evaporation and CO₂ desorption as verified by FT-IR (results presented below). From 411–543 K, the weight loss is 24.67% and this may be attributed to the decomposition of PILs aggregated within the hexagonal pore path. The weight loss is 28.01% in the third stage, which is believed to be the decomposition of PILs adsorbed in silica rings, as well as on silicon wall surface via hydrogen bond effects. Thus, the totally weight loss of PILs in S15-66[TEPA] is 52.68%, which is in good agreement to the elemental analysis data of 55.12% (presented below).

3.1.4. FT-IR analysis

The FT-IR spectra and absorption peaks for the bare support, S15-66[TEPA], and S15-66[TEPA]-CO₂ are depicted in Fig. 4 and

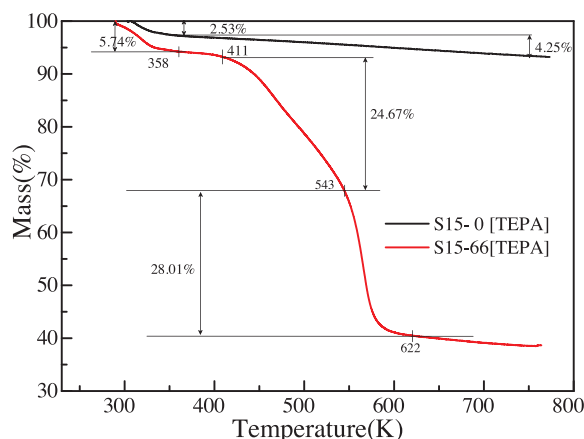


Fig. 3. TGA curves of S15-0[TEPA] and S15-66[TEPA]. Thermogravimetric analysis only for the 66 wt% loading composition due to its best performance.

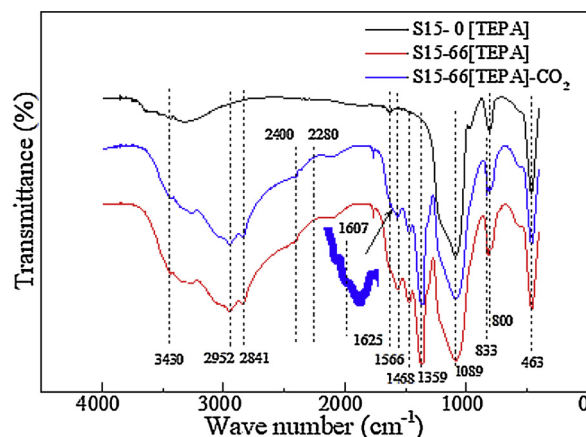


Fig. 4. FT-IR spectra and absorption peaks of bare support, S15-66[TEPA], and S15-66[TEPA]-CO₂.

Table 2

Assignment of FT-IR absorption peaks.

Absorption peaks (cm ⁻¹)	Assignment	Species
3430	Si-OH and H-O-H stretching vibration	SBA-15: Si-OH, physisorbed water
2820-2970	C-H stretching vibration	PIL: CH ₂
2400-2280	CO ₂ stretching vibration	Gas phase CO ₂
1625	H-O-H stretching vibration	physisorbed water
1607	COO ⁻ stretching vibration	NHCOO ⁻
1566	N-H bending vibration	PIL: NH ₂
1468	C-H deformation vibration	PIL: CH ₂
1359	NO ₃ ⁻ vibration	PIL: NO ₃ ⁻
1089	Si-O-Si asymmetric stretching vibration	SBA-15: Si-O-Si
833	NO ₃ ⁻ vibration	PIL: NO ₃ ⁻
800	Si-O-Si symmetric stretching vibration	SBA-15: Si-O-Si
463	Si-O-Si bending vibration	SBA-15: Si-O-Si

summarized in Table 2. Through a comparison of the characteristic absorption peaks for the bare support and S15-66[TEPA], it is readily apparent that the PIL was successfully impregnated into the support. In the case of the bare support, the strong characteristic absorption bands at 1089, 800, and 463 cm⁻¹ are assigned to asymmetric and symmetric stretching vibration peaks, and the bending vibration peak of Si-O-Si bonds, respectively [38]. In addition, the peak at 3430 cm⁻¹ is consistent with the Si-OH and H₂O absorption bands region of 3600–3200 cm⁻¹. The presence of this peak indicates the existence of silanol groups and adsorbed water molecules on the silicon walls. The presence of physisorbed water also can be confirmed by the stretching vibration peak at 1625 cm⁻¹ [39]. After [TEPA][NO₃] was impregnated, the emerging characteristic absorption peaks for the functionalized sorbent are at 2952, 2841, 2400, 2280, 1566, 1468, 1359, and 833 cm⁻¹. The absorption bands between 2820 cm⁻¹ and 2970 cm⁻¹ are stretching vibrations of the C-H bond in CH₂ groups and the peak observed at 1468 cm⁻¹ is attributed to the deformation vibration of the C-H bond [40]. In addition, the peak at 1566 cm⁻¹ corresponds to bending vibration peaks of N-H from the primary amines in the PIL [41]. Moreover, the very strong characteristic absorption peak at 1359 cm⁻¹ and moderate absorption peak at 833 cm⁻¹ are assigned to the vibrations of NO₃⁻ groups. Therefore, it can be confirmed that the PIL has been impregnated into SBA-15 successfully. Finally, the absorption bands from 2400 cm⁻¹ to 2280 cm⁻¹ are related to the stretching vibration of CO₂ molecules in the gas phase following physisorption.

The infrared spectrum of S15-66[TEPA]-CO₂, which represents S15-66[TEPA] following a CO₂ adsorption experiment, indicates that CO₂

Table 3

Comparison of elemental composition and actual PIL loading of S15-x[TEPA].

Sample	N (%)	C (%)	H (%)	N content (mmol/g)	Actual PILs loading (%)
S15-0 [TEPA]	0.07	0.07	0.60	0.05	0.00
S15-33[TEPA]	8.34	10.56	3.49	5.96	25.05
S15-50[TEPA]	12.48	15.73	4.90	8.91	37.49
S15-66[TEPA]	18.35	22.71	6.32	13.11	55.12
S15-75[TEPA]	21.01	26.25	7.21	15.00	63.09
S15-80[TEPA]	22.94	28.68	7.69	16.39	68.90
S15-83[TEPA]	23.34	29.27	8.07	16.67	70.09
S15-86[TEPA]	24.95	31.32	7.98	17.82	74.94

are captured through interactions with the amine groups in the PILs, forming carbamate. When the spectrum for S15-66[TEPA]-CO₂ is compared with the spectrum of S15-66[TEPA], there is an emerging characteristic absorption peak at 1607 cm⁻¹. The peak is associated with stretching vibration of COO⁻ in NHCOO⁻ groups [42]. The emergence of this peak demonstrates that the nature of CO₂ adsorption is chemisorption and resulting in the formation of carbamate moieties. This was further confirmed by the isosteric heat data and DFT calculation results below.

3.1.5. Element analysis

The elemental composition of S15-x[TEPA] were evaluated to understand the actual loading of PILs in the support. Table 3. shows that as the ratio of the PIL to SBA-15 increases, the nitrogen contents of the hybrid sorbents had a corresponding increase from 8.34% to 24.95%. The actual PIL loading of all samples is also listed in Table 3, and the results demonstrate the loading levels are lower than the theoretical weight percentages due to the incomplete impregnation as compared to the reaction mixture.

3.2. Adsorption behaviors

3.2.1. Effects of PIL loading level and adsorption temperature

The CO₂ breakthrough curves and adsorption capacities of S15-x [TEPA] at 313 K and 0.15 bar are shown in Fig. 5a and b. As a control, the adsorption capacity of SBA-15 was measured as only 0.24 mmol/g. This capacity is due to weak physisorption of CO₂ on the surface hydroxyl groups. With increasing PIL loading, the breakthrough time and CO₂ adsorption capacities of the sorbents raise due to presence of more active sites. However, when the PIL loading content exceeds 66 wt%, the breakthrough time and CO₂ adsorption capacities quickly decline. This result is attributed to blockage of the pores and subsequent loss of internal surface area from excess PIL. Similar results were also observed in our previous studies [37]. The maximum adsorption capacity measured in this study was 1.88 mmol/g for S15-66[TEPA].

A comparison of CO₂ adsorption capacities for various IL functionalized materials are summarized in Table 4. It shows that the CO₂ adsorption capacity of S15-66[TEPA] measured here is superior to many supported IL-functionalized systems previously reported in literature [36,43,44]. Moreover, this system exhibits an excellent performance under low CO₂ partial pressures as well.

The temperature of industrial flue gas usually falls in the range of 318 K and 348 K. Therefore, it is necessary to evaluate the effect of temperature on the adsorption capacity of sorbents. Fig. 5c and d exhibit that as the operating temperature increases, the CO₂ breakthrough time and adsorption capacity of S15-66[TEPA] first increases, before slightly declining at temperatures above 333 K. The maximum adsorption capacity of S15-66[TEPA] is 2.15 mmol/g at 333 K. The activity of binding sites in the PIL and diffusion of CO₂ to the active sites play dominant roles in the adsorption performance of sorbents. With increasing temperatures, the interaction between amine groups and CO₂ are enhanced, thus more CO₂ adsorbs with the amine groups. However,

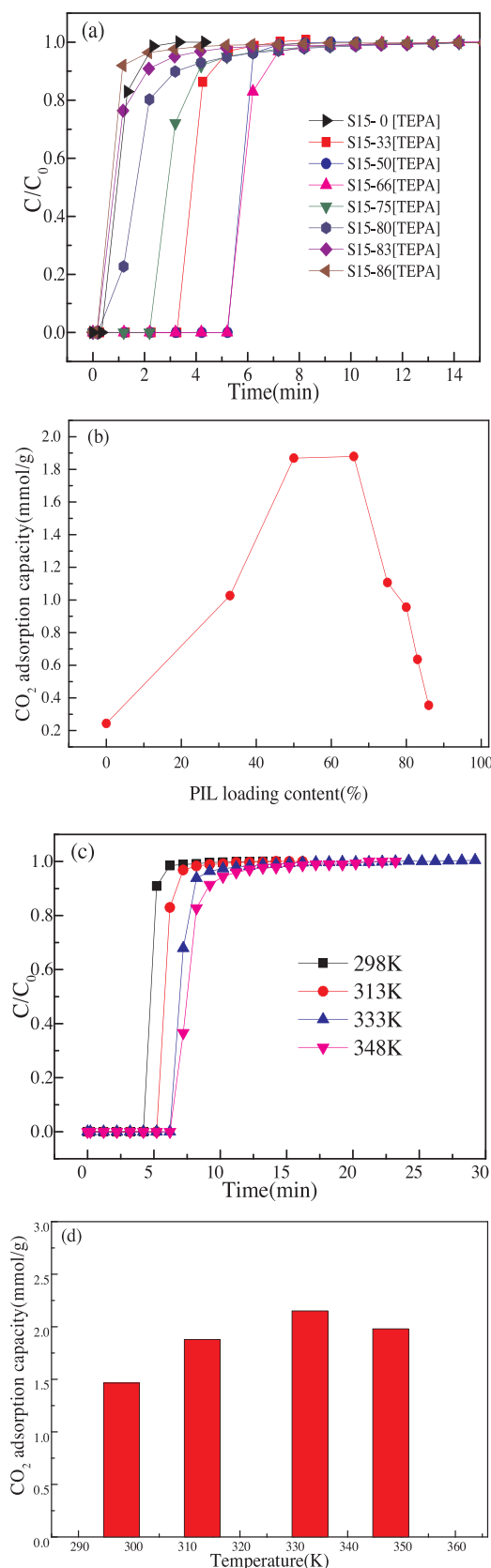


Fig. 5. Representative CO₂ breakthrough curves (a) and adsorption capacity (b) of S15-x[TEPA] at 313 K and 0.15 bar. Representative CO₂ breakthrough curves (c) and adsorption capacity (d) of S15-66[TEPA] at different temperatures.

Table 4
Comparison of CO₂ adsorption capacities of ILs functionalized materials.

Support	Functionalization	Ionic liquid & content	Adsorption capacity (mmol/g)	Experimental conditions		Ref.
				P _{CO₂} (bar)	T (°C)	
Silica particles (0.5–1.5 nm)	grafting	[P8883]TFSI	0.99	1	40	[45]
Silica particles (0.5–1.5 nm)	Impregnation	[Omim]TFSI(20%)	0.62	1	40	[45]
Silica particles (0.5–1.5 nm)	Impregnation	[Emim]TFSI(20%)	0.24	1	40	[45]
MCM-41	grafting	ILCLM50	1.48	1	25	[46]
MCM-41	grafting	ILBF ₆ M50	0.34	1	25	[46]
MCM-41	grafting	ILPF ₆ M50	0.57	1	25	[46]
MCM-41	grafting	ILTf ₂ NM50	0.23	1	25	[46]
MCM-41	Impregnation	[P4444][3-aminopyrazole](67%)	1.84	1	20	[47]
MCM-41	Impregnation	[P4444][imidazole](67%)	0.91	1	20	[47]
p-Al ₂ O ₃	Impregnation	[P4444][3-aminopyrazole](50%)	1.2	1	20	[47]
p-Al ₂ O ₃	Impregnation	[P4444][imidazole](50%)	1.16	1	20	[47]
α-zirconium phosphate	Impregnation	BMIMCl (70%)	0.31	1	30	[48]
montmorillonite	Impregnation	BMIMCl (24%)	0.4	1	30	[48]
MCM-41	Impregnation	[P66614][2-Op](50%)	1.17	1	20	[36]
MCM-41	Impregnation	[P66614][2-Op](50%)	0.56	0.4	20	[36]
Silica beads	Impregnation	EZT3(25%)	1.98	1	40	[20]
SBA-15	Impregnation	EZT3(50%)	4.7	1	40	[27]
PMMA	Impregnation	[EMIM][Lys](50%)	1.38	0.5	30	[43]
MCM-41	Impregnation	[C4PYR] + [TF2N] – (42%)	1.25	1	25	[49]
γ-Al ₂ O ₃	Impregnation	[Emim][Gly](16%)	0.58	0.15	30	[44]
SBA-15	Impregnation	[TEPA][NO ₃](66%)	2.15	0.15	60	This work

theoretically the CO₂ capture by amine groups is an exothermic reaction. This suggests that the reaction is controlled by CO₂ diffusion under lower temperature. When the thermodynamics and dynamics achieve balance at 333 K, the sorbent reaches the maximum CO₂ adsorption capacity. When the adsorption temperatures rises above 333 K, the reaction is controlled by thermodynamics, and the CO₂ desorption rate is faster than adsorption rate leading to a decrease in adsorption capacity.

3.2.2. Adsorption kinetics

Gas adsorption rates for sorbents are determined by surface properties and intra-particle diffusion. In order to obtain insights into the adsorption kinetics and rate-controlling step of adsorption, the CO₂ dynamic uptake was plotted against $t^{0.5}$ using the intra-particle diffusion model (IPDM) proposed by Weber and Morris [50]:

$$q_t = kt^{0.5} + C \quad (3)$$

Where, q_t (mmol g⁻¹) represents CO₂ uptake at time t (s); k (mmol g⁻¹ s^{-0.5}) is the intra-particle diffusion rate constant; and C (mmol g⁻¹) is related to the intercept. The model contains three distinct steps: (I) the initial stage, where the external surface adsorption or boundary layer diffusion is dominant; (II) the splitting adsorption region, which is controlled by intra-particle diffusion; and (III) the final equilibrium step. The slope of the Weber–Morris plot of q_t versus $t^{0.5}$ (s^{0.5}) in the time intervals I, II, and III are denoted as k_I , k_{II} , and k_{III} .

Representative IPDM kinetic plots for the sorbents and the fitting value for k_{II} as a function of the PIL loading level are shown in Figs. 6a and b, respectively. In the case of the bare support, the sorption kinetics are best characterized by a 2-step process, which corresponds to instantaneous physical adsorption (k_I) and the final equilibrium step (k_{III}). In the physisorption stage, CO₂ is captured through H-bonding with surface hydroxyl groups forming silanol–CO₂ complexes [51]. In contrast, the hybrid sorbents show a 3-step mechanism and the rate constants for each step are listed in Table 5. It is readily apparent that k_{II} increases with the PIL loading level up to 66 wt%. This increase is associated with the availability of active sites and the strong affinity of the PIL functional groups towards CO₂, which outweighs the negative effect of impregnation on the structural properties. Further increases in the PIL loading level beyond 66 wt% reintroduces diffusion limitations due to excess PILs, as demonstrated by lower k_{II} values. The third step involves the physisorption-dominated equilibrium process where the intra-particle diffusion starts to slow down due to the saturation [52].

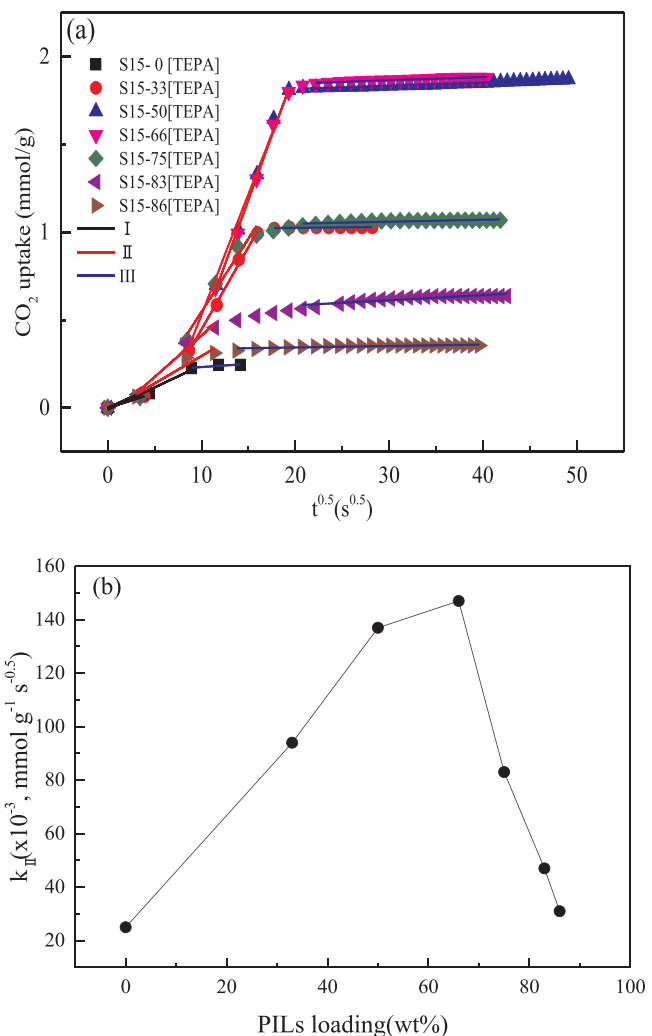


Fig. 6. Representative IPDM kinetic plots of S15-x[TEPA] (a) and correlation plots of k_{II} as a function of PIL loading (b).

Table 5

Comparison of diffusion rate constants from the IPDM model for each stage of adsorption.

Samples	IPDM(mmol g ⁻¹ s ^{-0.5})			Ref.
	k _I	k _{II}	k _{III}	
S15-0 [TEPA]	25·10 ⁻³	–	4·10 ⁻³	
S15-33[TEPA]	17·10 ⁻³	94·10 ⁻³	1·10 ⁻³	
S15-50[TEPA]	19·10 ⁻³	137·10 ⁻³	2·10 ⁻³	
S15-66[TEPA]	18·10 ⁻³	147·10 ⁻³	2·10 ⁻³	
S15-75[TEPA]	18·10 ⁻³	83·10 ⁻³	1·10 ⁻³	
S15-83[TEPA]	24·10 ⁻³	47·10 ⁻³	3·10 ⁻³	
S15-86[TEPA]	21·10 ⁻³	31·10 ⁻³	1·10 ⁻³	
MCM-41-60DBUOH		15·10 ⁻³		[53]
S15-60DBUOH		31·10 ⁻³		[53]
S15-25EZT3		41·10 ⁻³		[27]
S15-50EZT3		50·10 ⁻³		[27]
S15-80EZT3		30·10 ⁻³		[27]
S15 (4.3) -50[P ₆₆₆₁₄][Triz]		29·10 ⁻³		[28]
SALG-AT-EZT3-25		25·10 ⁻³		[20]

Clearly, increasing the number of available adsorption sites and eliminating the viscosity limitation of bulk PILs for mass transfer are accomplished together by spreading the PIL across the pore surface of SBA-15. For all hybrid sorbents, the k_{II} is significantly higher than k_I and k_{III}, therefore the splitting adsorption process is the rate-controlling step.

The CO₂ adsorption nature of previously reported IL-functionalized [20,27,28] and amine-modified support systems [53] representing both chemisorption and physisorption approaches have been previous reported. The CO₂ uptake profiles for these systems, under similar conditions, have also been analyzed with the IPDM model, while demonstrating 3-step sorption kinetic mechanisms. The previously published values for k_{II} in these systems are summarized in Table 5. In this table it is clearly evident that the rate limiting step for S15-66[TEPA], $147 \times 10^{-3} \text{ mmol g}^{-1} \text{ s}^{-0.5}$, is at least threefold higher than related sorbent systems.

3.3. Adsorption isotherms and isosteric heat

According to the FT-IR analysis, CO₂ is adsorbed through two different mechanisms: chemisorption and physisorption. Thus, the dual-site Langmuir model [54,55] can be used to describe the adsorption behavior of CO₂ in S15-66[TEPA]. The equation is defined as:

$$q_e = \sum_{j=1}^N q_{m,j} \frac{b_j P}{1 + b_j P} \quad (4)$$

Where, q_m and b_j are Langmuir parameters representing the maximum adsorption capacity and adsorption affinity constant for site j , respectively; q_e represents the equilibrium adsorption capacity; P refers to the pressure of CO₂.

Further, the isosteric heat of adsorption (ΔH_{iso}) is commonly used to describe the enthalpy change during an adsorption process. This is calculated from adsorption data at different temperatures using the Clausius-Clapeyron equation as follows [56,57]:

$$\Delta H_{\text{iso}} = R \frac{d \ln p}{d \left(\frac{1}{T} \right)} \quad (5)$$

Where, p is the pressure (bar) at equilibrium uptake; T represents the temperature (K); R refers to the universal gas constant, 8.3145 J/(mol·K).

The experimentally obtained CO₂ adsorption isotherms, dual-site Langmuir model fits, and isosteric heat of adsorption for CO₂ on S15-66[TEPA] are shown in Fig. 7. The isotherms present sharp increases at pressure < 0.15 bar and much slower increases at pressures from 0.15 to 1.0 bar. In addition, the dual-site Langmuir model closely fits the

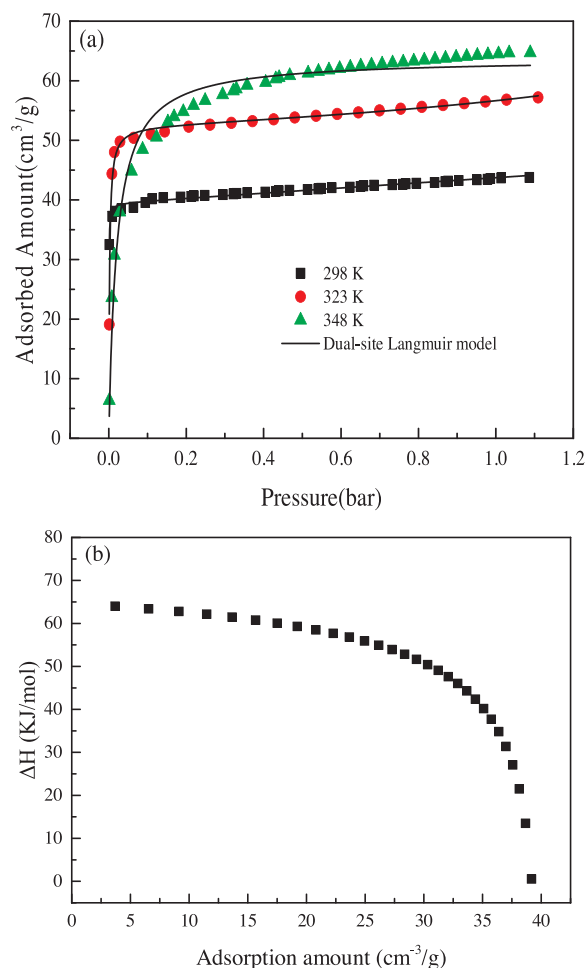


Fig. 7. (a) CO₂ adsorption isotherms and dual-site Langmuir model fits for S15-66[TEPA] at 298 K, 323 K and 348 K. (b) Isosteric heat of adsorption for CO₂ on S15-66[TEPA].

experimental data across this range of pressures, for the different experimental temperatures. These results demonstrate the heterogeneous nature of the surface [58].

The maximum isosteric heat of adsorption for CO₂ on S15-66[TEPA] is 64.00 KJ/mol. This value is significantly higher than 40 KJ/mol, which demonstrates that the CO₂ adsorption on these sorbents is dominated by chemisorption [59]. Fig. 7b shows that the isosteric heat of adsorption plot decreases slowly for adsorption levels < 32 cm³/g, followed by a sharp decrease. This is attributed to the strong interaction between CO₂ and the S15-66[TEPA] sorbent at low surface coverage, resulting in high heat of adsorption. Furthermore, as the surface coverage increases, the amine active sites become occupied by CO₂ and later adsorption is dependent on physisorption [60].

3.4. CO₂ adsorption mechanism

[TEPA][NO₃] has multiple amine groups available for CO₂ binding. In order to understand the interactions between amine groups and CO₂ at the molecular level, DFT simulations were conducted to calculate the binding energies for CO₂ adsorption over different amine groups. Compared with the total energy of the structures that NO₃⁻ binding with five different amine sites, the energy minimized structure and molecular dimensions of [TEPA][NO₃] are shown in Fig. 8. In this figure it can be seen that the central nitrogen of TEPA is protonated by the addition of nitric acid to form a quaternary ammonium center in the cation. The structure of these contact-ion pairs is similar to ethylammonium nitrate as reported by Emel'yanenko et al. [61].

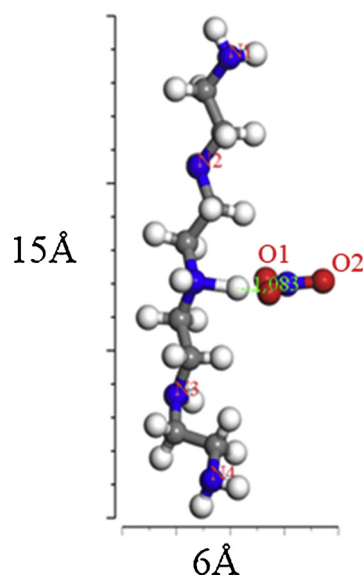


Fig. 8. Energy minimized structure of [TEPA][NO₃]. Color code: gray, C; red, O; blue, N; white, H.

Table 6
Binding energies calculated at the B3LYP/DNP level.

Structure	BE B3LYP/DNP (kJ/mol)
[TEPA][NO ₃](N1)-CO ₂	−23.0
[TEPA][NO ₃](N2)-CO ₂	−29.6
[TEPA][NO ₃](N3)-CO ₂	−28.4
[TEPA][NO ₃](N4)-CO ₂	−23.9
[TEPA][NO ₃](O1)-CO ₂	−19.0
[TEPA][NO ₃](O2)-CO ₂	−16.3
Carbamate-N(1)H ₃ ⁺ -[TEPA][NO ₃] ₂	−42.6
Carbamate-N(2)H ₂ ⁺ -[TEPA][NO ₃] ₂	−64.0
Carbamate-N(3)H ₂ ⁺ -[TEPA][NO ₃] ₂	0.6
Carbamate-N(4)H ₃ ⁺ -[TEPA][NO ₃] ₂	−54.9

The calculated BE of carbamate-N(2)H₂⁺-[TEPA][NO₃]₂ is −64.0 kJ/mol, which is consistent with other theoretical calculations [62] and the experimental results discussed above. The strategy for understanding the mechanisms of CO₂ capture by amine groups in the PILs involved the simulation of the potential carbamate-PIL structures. First, for all of the systems investigated, CO₂ molecules are weakly adsorbed to amine groups and nitrates (denoted as [TEPA][NO₃](species) – CO₂), leading to the relaxation of specific N and O sites in the PIL structure and the C site of CO₂. Then, the amine active site produces a zwitterionic carbamate anion that is stabilized by a hydrogen bond with a protonated amine site in a neighboring PIL. When the BE for each

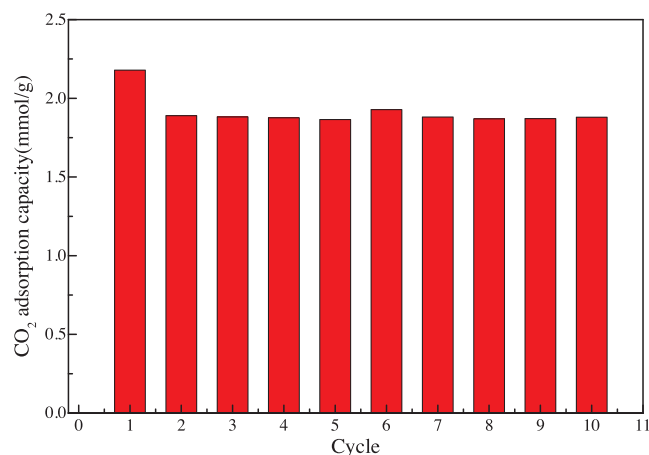
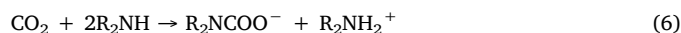


Fig. 10. Cyclic regeneration of S15-66[TEPA].

nitrogen group are compared, as seen in Table 6, it can be seen that CO₂ preferentially interacts with the secondary amine -N(2)H to form carbamate based on this interaction having the highest BE. This binding interaction contains intermolecular hydrogen bonds of N–HON[−]OO[−] and N–H–NH₂. These results provide a theoretical basis for the design of highly efficient CO₂ adsorbents. Furthermore, the optimal interaction configuration is presented in Fig. 9. The reaction can be depicted as followed:



3.5. Adsorbent regenerability

The long-term operating performance of S15-66[TEPA] was evaluated through cyclic adsorption/regeneration experiments and the results in Fig. 10 show that nearly 90% of the original CO₂ capture capacities are reversible. The sorbent underwent CO₂ adsorption at 333 K and 0.15 bar, and then desorption occurred under a N₂ steam at 373 K for 30 min. over 10 identical cycles. Fig. 10 shows that the adsorption capacity of S15-66[TEPA] decreased by 13.3% after the first cycle. This decrease can be attributed to the strong diffusion limitations generated by carbamates, which inhibit the complete desorption of CO₂ molecules under the regeneration conditions tested [63]. However, there are only minor fluctuations in the adsorption capacity over the subsequent 9 cycles. Therefore, S15-66[TEPA] has excellent regeneration capacity.

4. Conclusions

A series of supported protic ionic liquid sorbents based on low-cost polyamine-based [TEPA][NO₃] confined into SBA-15 have been synthesized, characterized, and evaluated for their CO₂ adsorption. The

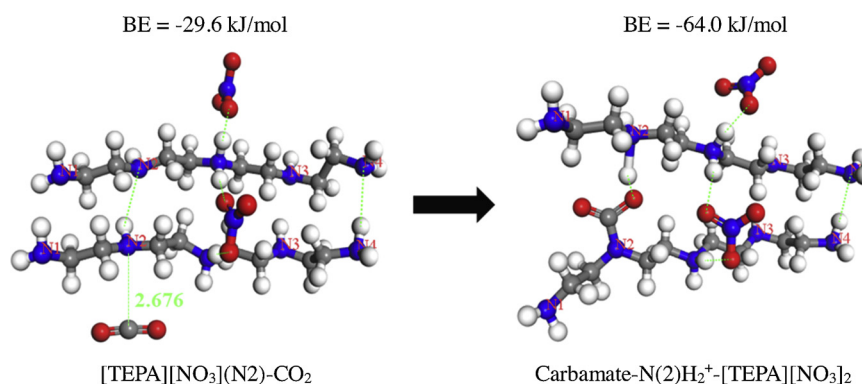


Fig. 9. Optimal interaction configuration of CO₂ over the -N(2)H in [TEPA][NO₃]. Color code: gray, C; red, O; blue, N; white, H.

S15-66[TEPA] system exhibited excellent CO₂ adsorption rates, high adsorption capacity, and regeneration. The adsorption rate in the sorbent rate-controlling step is $147 \times 10^{-3} \text{ mmol g}^{-1} \text{ s}^{-0.5}$, which is at least two times higher than any previously reported IL-functionalized or amine-modified support systems. In addition, the hybrid sorbent showed great CO₂ uptake capacity of 2.15 mmol/g at 333 K and 0.15 bar, which represents an excellent performance under low CO₂ partial pressures when compared with other IL-functionalized sorbents. According to the adsorption isotherms and isosteric heat of adsorption, the CO₂ adsorption process involved both physisorption and chemisorption. DFT calculations suggested that CO₂ molecules preferentially react with the secondary amine, -N(2)H, in [TEPA][NO₃] to form carbamate. Finally, after 10 adsorption/desorption cycles, the CO₂ adsorption capacity of the sorbent maintained 90% of the first cycle capacity. In summary, [TEPA][NO₃] impregnated SBA-15 sorbents provide a low-cost, high adsorption capacity alternative with fast adsorption kinetics, and regenerability. The high efficiency can be explained from the synergistic effects of strong binding affinity between the PILs and CO₂ and fast intra-particle diffusion rates. Therefore S15-66[TEPA] represent a promising candidate for CO₂ capture in industrial applications.

Declaration of Competing Interest

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.jcou.2019.08.012>.

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